

BABES-BOLYAI UNIVERSITY ▪ FACULTY OF MATHEMATICS AND COMPUTER SCIENCE

ARTIFICIAL INTELLIGENCE SPECIALISATION

Neural Decoding of Visual Perception from fMRI

CLIP-Guided Retrieval and Diffusion-Based Reconstruction

86.3% CSLS Recall@1 on SHARED1000 ▪ 95 % CI [84.0, 88.4]

Iepure Antoniu

supervised by Prof. univ. dr. Anca-Mirela Andreica

Cluj-Napoca ▪ June 2026

The problem: decoding what the brain sees

Thesis ■ Chapter 1.1, Motivation & Formulation

Question. Given the fMRI response of a subject viewing a natural image, can we recover *which* image was perceived?

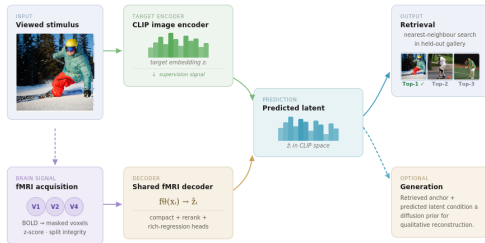
Approach. Map brain activity into a **CLIP-aligned latent space** rather than to pixels:

$$\hat{z}_i = f_{\theta}(x_i), \quad f_{\theta} : \mathbb{R}^V \rightarrow \mathbb{R}^d$$

$V \approx 15,700$ voxels ■ $d = 768$ (CLIP ViT-L/14)

Guiding hypothesis

Held-out retrieval is limited less by the fidelity of any single predicted embedding



Single-example decoding path — NSD 7T, subj01,
nsdgeneral ROI (Figure 1.2 in thesis)

Why predict CLIP embeddings, not pixels?

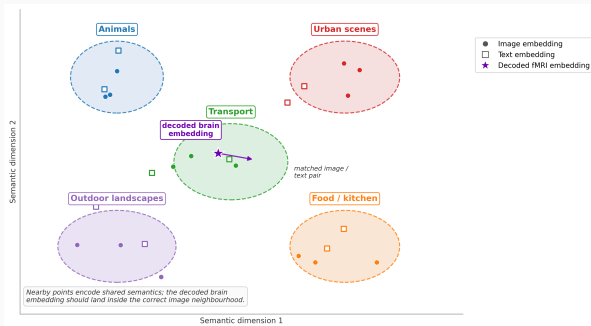
Thesis ■ Chapter 2.5, CLIP as the Target Representation

CLIP (Radford et al., 2021) learns a joint image–text space where **cosine similarity tracks semantic similarity**.

For brain decoding this gives us:

- a **stable, low-dim target** ($d=768$) instead of fragile pixels;
- nearest-neighbour **retrieval** as a clean evaluation endpoint;
- a **controllable handle** for diffusion-based reconstruction.

The pipeline is *retrieval-first*: judged by

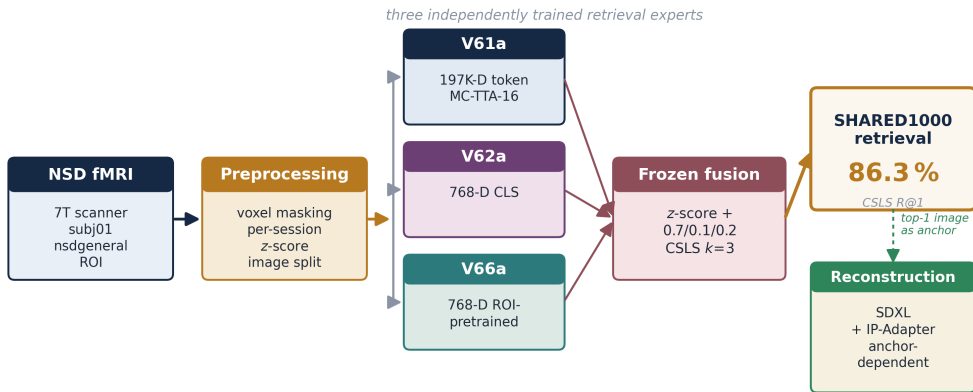


Shared CLIP embedding space — the decoded fMRI embedding

should land in the correct semantic neighbourhood (Figure 2.4 in thesis)

From fMRI to retrieval — the full pipeline

Thesis ■ Chapter 3, Methodology (Figure 1.1 in the thesis)



shared preprocessing → three independent experts → frozen score-level fusion → held-out retrieval endpoint

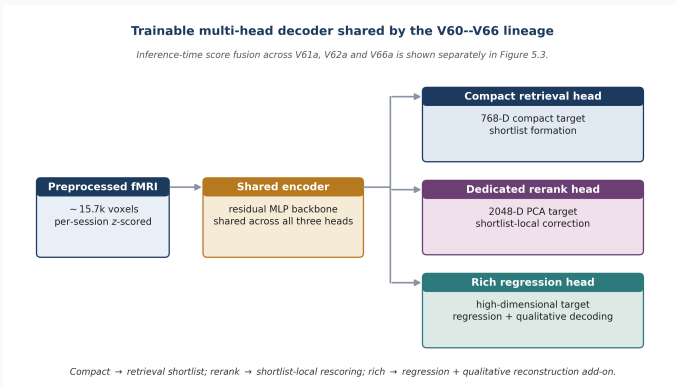
One encoder, three roles

Thesis ■ Chapter 3.5, Architecture Evolution (Figure 3.2 in the thesis)

- **Compact** — global retrieval
- **Rerank** — shortlist-local correction
- **Rich** — reconstruction-oriented regression

Design principle.

Do not force one latent space to solve all objectives.



Main contribution: frozen score-level triple fusion

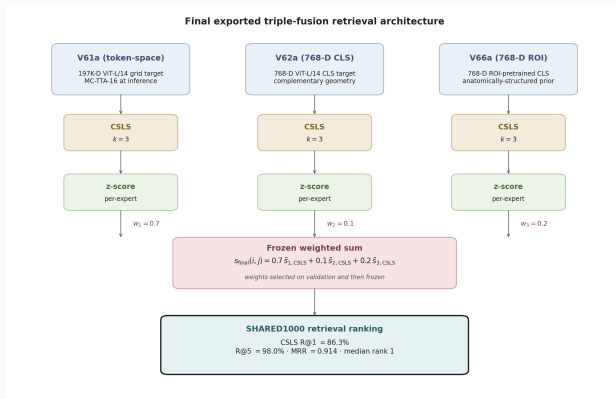
Thesis ■ Chapter 3.6, Inference & Score Fusion (Equation 3.11)

Claim. A frozen weighted sum of hubness-corrected, z-score-normalised scores from **three independent** retrieval experts, applied **only at inference**, is a defensible retrieval policy.

Each expert: CSLS-corrected at $k=3$.

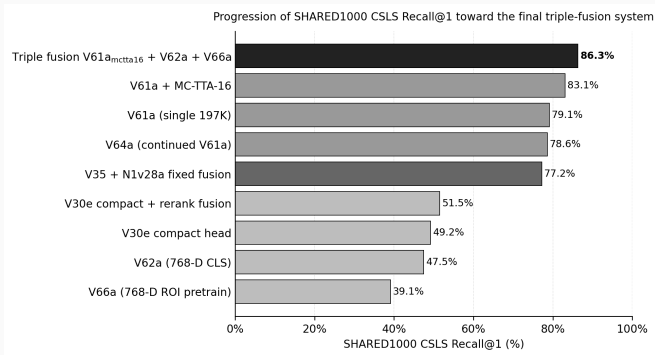
$$s_{\text{final}}(i, j) = \sum_{e=1}^3 w_e \tilde{s}_e(i, j)$$

weights **0.7/0.1/0.2**, chosen on validation, then **frozen** before SHARED1000.



Headline result — SHARED1000 retrieval

Thesis ■ Chapter 5.4, Final Retrieval Results (Table 5.1)



86.3%

CSLS R@1 on
SHARED1000

95 % CI [84.0, 88.4] ■

Clopper–Pearson

R@5 = **98.0%** ■ MRR = **0.914** ■

median rank = **1**

+3.2 **pp** over strongest single expert

+9.1 **pp** over V35 + N1v28a
milestone

McNemar $p = 3.3 \times 10^{-6}$

Why the +3.2 pp gain: aggregate complementarity

Thesis ■ Chapter 5.5, Per-trial paired analysis (Table 5.4)

Paired transition (V61a-MC-TTA vs. triple fusion):

	Fusion R@1	Fusion wrong
V61a R@1	823	8
V61a wrong	40	129

Net gain = $40 - 8 = +32$ trials = +3.2 pp.

Exact McNemar $p = 3.3 \times 10^{-6}$.

Aggregate claim: the paired analysis is *consistent with complementary correction*.

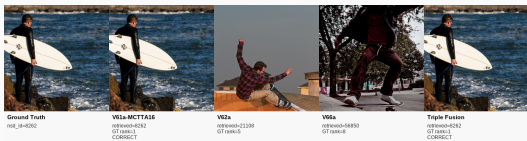
Expert roles

- **V61a** — 197 376-D token, MC-TTA-16; strongest single ranker (83.1 %).
- **V62a** — 768-D CLS; complementary score geometry.
- **V66a** — 768-D ROI-pretrained; structurally different prior.

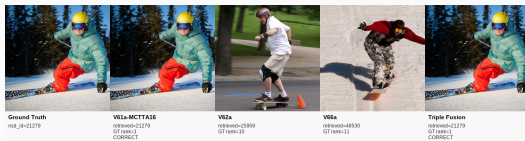
Auxiliary experts are useful because their score geometry is **complementary** — not because they are strong alone.

Qualitative retrieval evidence

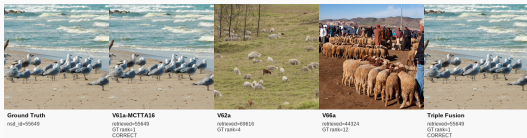
Thesis ■ Chapter 5.6, Qualitative Retrieval (Figures 5.5–5.6)



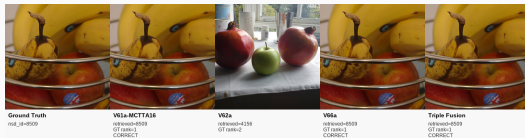
NSD 8262 — *surfer* (fusion rank 1)



NSD 21279 — *skier* (fusion rank 1)



NSD 55649 — *seagulls* (fusion rank 1)



NSD 8509 — *fruit bowl* (fusion rank 1)

Each row: ground truth | V61a-MCTTA | V62a | V66a | **triple fusion** — rank of true stimulus annotated.

Illustrative rank-1 preservation examples; paired rescue/break evidence is reported separately in 8/10
Table 5.4.

Secondary add-on: anchor-dependent reconstruction

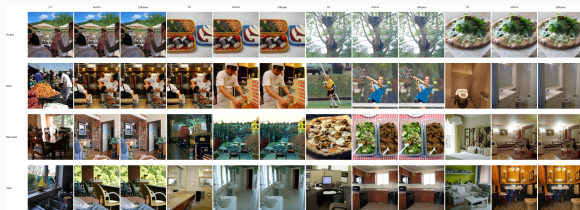
Thesis ■ Chapter 5.8, Reconstruction Add-on (Table 5.8)

Retrieval is the thesis benchmark; reconstruction is a controlled downstream add-on.

Pipeline — triple-fusion top-1 as **anchor** → SDXL 1.0 + IP-Adapter; fMRI-predicted CLIP injected; best-of-16.

Metric ($n=141$)	Retr.	Diff.
SSIM	0.202	0.214
PSNR (dB)	9.54	9.60
CLIP-I sim	0.744	0.750
AlexNet(5) 2-way	—	86.2 %

Scope: anchor-dependent.



Contributions, limitations, outlook

Thesis ■ Chapter 6, Conclusions and Future Work

Contributions

- Reproducible
NSD→CLIP retrieval
pipeline
- *Compact / rerank / rich*
role separation
- Frozen triple fusion:
86.3 % CSLS R@1

Limitations

- Single-subject protocol
(subj01)
- Reconstruction is
anchor-dependent
- Not protocol-comparable
to MindEye / MindEye2

Future Work

- Multi-subject scaling
- Reconstruction ablations
- Learned
uncertainty-gated fusion

— TRANSITION —

Application demo next

A short live demonstration (≈ 5 min)

*I will now demonstrate the application and show how the reported retrieval result is connected to **reproducible artifacts and per-trial outputs**.*

86.3% CSLS R@1 ▪ 95% CI [84.0, 88.4] ▪ McNemar $p < 10^{-5}$

(Q & A after the demo)

Iepure Antoniu

supervised by Prof. univ. dr. Anca-Mirela Andreica ▪ UBB FMI ▪ June 2026

Backup: reproducibility & demo evidence

Thesis ■ *Chapter 4 + Table 4.1, artifact map*

Claim	Artifact
Per-expert configs	configs/experiments/{V61a,V62a,V66a}*.yaml
Fusion result (86.3 %)	experimental_results/fusion_triple_86/ triple_fusion_shared1000_86.json
Per-trial fused ranks	experimental_results/fusion_triple_86/ per_trial_fused_ranks.csv
Reconstruction metrics	experimental_results/reconstruction_sdxl_ipa/ metrics_summary.json
Qualitative panels	thesis/v62_thesis/query_{8262,21279,55649,8509}.png
V62a manifold report	reports/manifold_analysis/20260514_V62a_*/

The application is not a separate demo layer — it reads the same exported artifacts used to report the thesis results.

Backup: full per-expert numbers

Thesis ■ *Chapter 5.4, Table 5.1*

System	Raw R@1	CSLS R@1	R@5	MRR	Med. rank	Weight
V30e (compact)	—	49.2 %	—	—	—	—
V30e + rerank	—	51.5 %	—	—	—	—
V35 + N1v28a (2-expert)	—	77.2 %	—	—	—	—
V61a (single, 197K)	66.0 %	79.1 %	—	—	—	—
V61a + MC-TTA-16	67.3 %	83.1 %	96.9 %	0.892	1	0.7
V62a (768-D CLS)	39.9 %	47.5 %	80.3 %	0.621	2	0.1
V66a (768-D ROI)	35.5 %	39.1 %	73.8 %	0.545	2	0.2
Triple fusion	72.1 %	86.3 %	98.0 %	0.914	1	frozen

Backup: key mathematical formulations

Thesis ■ Chapter 3.4 + 2.7 + 3.6

InfoNCE (compact-head retrieval objective):

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(\hat{z}_i, z_i) / \tau)}{\sum_{j=1}^N \exp(\cos(\hat{z}_i, z_j) / \tau)}$$

CSLS hubness correction (Conneau et al., 2018):

$$\text{CSLS}(x, y) = 2 \cos(x, y) - r_T(x) - r_S(y), \quad r_T(x) = \frac{1}{k} \sum_{y' \in \mathcal{N}_k(x)} \cos(x, y')$$

Frozen triple-fusion score (inference-time only):

$$s_{\text{final}}(i, j) = 0.7 \tilde{s}_{\text{V61a}}(i, j) + 0.1 \tilde{s}_{\text{V62a}}(i, j) + 0.2 \tilde{s}_{\text{V66a}}(i, j)$$

where $\tilde{s}_e = (s_e - \mu_e) / \sigma_e$ is per-expert z-score normalised CSLS at $k=3$.

Backup: fusion-level hubness drop

Thesis ■ Chapter 5.7, Fusion-level Hubness Analysis (Table 5.3)

The fused score matrix has lower hubness than *any* of its three constituents — a genuine system-level property.

System	V61a	V62a	V66a	Fusion
Cosine Gini	0.457	0.600	0.641	0.404
CSLS Gini	0.255	0.448	0.560	0.204
Max hub count	5	7	6	3

Why this matters

Hubness is the well-known failure mode of high-dimensional cosine retrieval: a small set of gallery items become nearest-neighbours of too many queries.

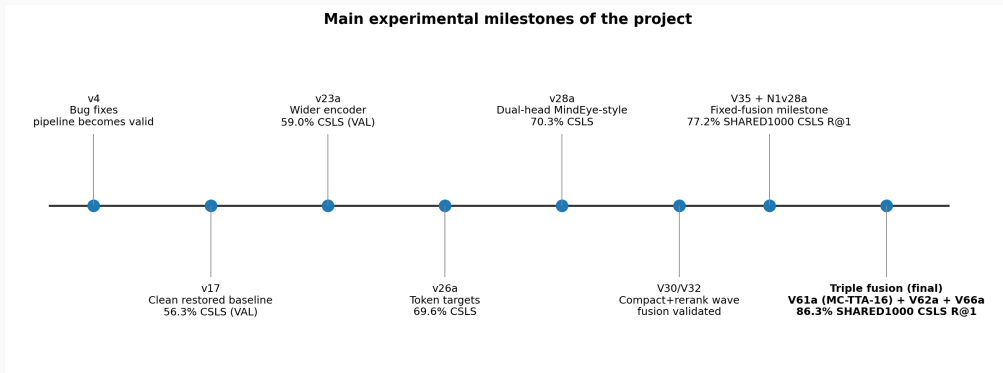
z-score normalisation + weighted combination disperse the score mass more evenly across the gallery.

No single image dominates the fused ranking. This is one of the mechanisms behind the **+3.2 pp** fusion gain.

CSLS neighbourhood $k=3$ ■ SHARED1000 target-embeddings protocol ■ 1000-image gallery.

Backup: experimental evolution timeline

Thesis ■ *Chapter 5.3, Qualitative Evolution (Figure 5.2)*



Eight design phases over the project lifecycle, from pipeline stabilisation to the frozen triple-fusion export.

Backup: literature context

Thesis ■ *Chapter 2.10, Representative Related Work*

Work	Approach	Relation to this thesis
Allen et al., 2022	NSD dataset (7T fMRI)	Benchmark and data source
Takagi & Nishimoto, 2023	SD-based reconstruction-first	Complementary paradigm
Ozcelik & VanRullen, 2023	Brain Diffuser	Latent-diffusion precedent
MindEye (Scotti, 2023)	Retrieval + reconstruction	Retrieval-first framing
MindEye2 (Scotti, 2024)	Shared-subject scaling	Multi-subject extension
This thesis	Frozen expert fusion	86.3 % R@1, single-subject